Sehat Saathi

Product Requirements Document

Teleconsultation Queue & Triage Management System for Rural Clinics

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1. Executive Summary

Sehat Saathi is an end-to-end, AI-powered teleconsultation and triage management platform built specifically for India's rural Primary Health Centres (PHCs). It addresses the chronic crisis of understaffing, unmanaged patient queues, fragmented health records, and the complete absence of an intelligent clinical triage layer in existing teleconsultation systems.

The platform is accessible to all patients regardless of literacy, device ownership, or internet connectivity — through voice, SMS, USSD, and a Progressive Web App. It serves two primary user groups: patients (through the Patient Portal) and doctors (through the Doctor Dashboard), with additional integrations for pharmacies, ASHA workers, and district health authorities.

Core capabilities include:

- Real-time community health heatmap for outbreak early detection
- Ayushman Card-linked digital health records with AI wellness recommendations
- Voice-based, multilingual symptom intake with NLP-powered triage
- SMS-based offline appointment booking and emergency ambulance dispatch
- Dynamic AI triage with 3-tier questionnaire and severity scoring
- Predictive queue management with full transparency for patients
- Pharmacy analytics dashboard with doctor-prescription integration
- Simplified audio diagnosis explanations for patient health literacy

This document outlines all product requirements, user stories, functional specifications, data models, API contracts, and implementation guidelines needed to build and ship Sehat Saathi v1.0.

2. Problem Statement

Rural India's healthcare infrastructure is under severe strain. The National Health Mission reports a critical shortage of doctors and paramedical staff at the PHC level, creating the following systemic failures:

Problem	Current State	Impact
Unmanaged Queues	No prioritisation exists — critical patients wait alongside minor ailments	Delayed care, preventable deaths
Doctor Shortage	1 doctor per 10,000+ patients in rural areas	Burnout, misdiagnosis, limited access
No Triage Layer	Teleconsult platforms route patients FIFO with no severity scoring	Inefficient resource use
Paper-Based Records	No longitudinal health data; patients carry physical files	Missed allergies, drug interactions
Language Barriers	Health interfaces only in English/Hindi	Exclusion of 600M+ dialect speakers
Connectivity Gaps	2G or zero data in many villages	Platform abandonment before registration
Medicine Gaps	Local pharmacy stock unknown to doctors or patients	Prescription non-compliance
Outbreak Blindness	No real-time disease surveillance at village level	Epidemics grow undetected

3. Goals & Non-Goals

3.1 Goals

- Reduce average patient wait time at rural PHCs by 60% through AI triage
- Enable 100% of rural patients to register and get triaged — regardless of literacy or device
- Provide doctors with structured, AI-generated pre-consultation patient briefs
- Enable early outbreak detection through real-time village-level health heatmaps
- Integrate pharmacy prescription fulfilment to reduce medication non-compliance
- Build patient trust through transparent, explainable queue management
- Deliver post-consultation diagnosis summaries in regional language audio
- Comply with ABHA (Ayushman Bharat Health Account) and NHM data standards

3.2 Non-Goals (v1.0)

- Full Electronic Health Record (EHR) system replacement — we integrate with ABHA, not replace it
- In-person physical examination capabilities
- Insurance claims processing (planned for v2.0)
- Mental health specialised triage (separate module, v2.0)
- Surgical / tertiary care referral management (v2.0)

4. User Personas

4.1 Patient — Raman Lal, 54, Farmer, Rampur Village

Literacy	Class 8 educated, speaks Bhojpuri, limited Hindi literacy
Device	Basic feature phone (no smartphone)
Connectivity	2G signal, no home WiFi
Health	Hypertension, Type 2 Diabetes, managed with monthly PHC visits
Pain Points	Travels 12 km to PHC, waits 3-4 hours, often told doctor is busy
Goals	Know his turn before leaving home, get medicines without another trip

4.2 Doctor — Dr. Priya Sharma, 31, MBBS, PHC Rampur

Load	Sees 80-120 patients per day with 1 assistant
Tools	Pen & paper registers, occasional WhatsApp consultation
Pain Points	No patient history available; spends first 5 min on basic intake; can't see next patient info while consulting
Goals	Pre-read patient summary before call, flag critical patients, reduce repeat consultations

4.3 Pharmacist — Suresh Gupta, Rampur Pharmacy

Context	Small generic pharmacy, 300 SKUs, no digital inventory
Pain Points	Last-minute medicine demands, unknown monthly consumption patterns, frequent stockouts
Goals	Receive prescriptions digitally, plan stock proactively, report reimbursements

4.4 ASHA Worker — Geeta Devi

Context	Covers 150 households, mobile phone with basic data plan
Role	Data entry for patients without devices, home visit scheduling
Goals	Register patients quickly, get alerts for high-risk households

4.5 District Health Officer (DHO)

Context	Monitors 50+ PHCs across the district
Goals	Early outbreak detection, resource allocation data, monthly PHC performance reports

5. System Architecture Overview

Sehat Saathi follows a microservices architecture with an offline-first design philosophy. The system is split into two primary portals sharing a common backend:

Layer	Technology	Purpose
Patient PWA	React Native Web / PWA	Runs on any smartphone or PC browser; offline via Service Workers
SMS Gateway	Twilio / Exotel IVR	Feature-phone users: register, book, and receive alerts via SMS/IVR
USSD Interface	Telecom USSD API (*HEALTH#)	Zero-data registration for non-smartphone users
AI Triage Engine	Python FastAPI + scikit-learn	Symptom scoring, severity classification, queue rank calculation
NLP Module	Custom Indic NLP + Google STT	Voice transcript → structured symptom JSON
Backend API	Node.js + Express	REST API gateway, auth, business logic
Queue Engine	Redis + BullMQ	Priority queue with real-time dynamic re-ordering
Database	PostgreSQL (primary) + CouchDB (sync)	Relational data + offline conflict-free sync
Notification Service	Firebase FCM + Twilio SMS	Push alerts, SMS alerts, voice callbacks
Maps / GIS	Google Maps Platform	Heatmap rendering, pharmacy geolocation
Pharmacy Module	Node.js microservice	Prescription routing, inventory API, analytics
Infra	AWS ap-south-1 (Mumbai) + Docker + k8s	HIPAA-adjacent compliant, auto-scaling

All patient data is stored encrypted at rest (AES-256) and in transit (TLS 1.3). The system integrates with ABHA APIs for health ID linkage and NHM APIs for government scheme reimbursement reporting.

6. Patient Portal — Detailed Feature Requirements

6.1 Community Health Heatmap System

The home screen of the Patient Portal features a dynamic village-level health heatmap that helps patients and authorities understand real-time disease trends across nearby villages.

6.1.1 Functional Requirements

- Display a Google Maps-based heatmap showing health activity levels across villages within a configurable radius (default: 30 km)
- Aggregate anonymised data from teleconsultation inputs, symptom questionnaires, ASHA worker reports, and pharmacy dispensing records
- Implement a 3-tier colour-coded alert system:

Colour	Zone	Trigger Condition	Action
GREEN	Safe	< 5% weekly symptom increase vs 4-week baseline	No action required; normal display
YELLOW	Early Warning	5–20% increase OR 3+ similar symptoms in 48 hrs	Alert sent to PHC doctor and ASHA worker
RED	Outbreak Alert	> 20% increase OR cluster of 5+ identical symptoms in 72 hrs	Automatic DHO notification + resource pre-positioning alert

- Support filtering by: disease/symptom type, time range (24h / 7d / 30d / custom), and geographic radius
- Update heatmap in near-real-time (every 15 minutes) using WebSocket push for online users; cached snapshot for low-bandwidth
- Detect and highlight clusters of: fever, respiratory illness, diarrhoea, skin infections, vector-borne diseases
- Generate automated alerts to District Health Officers when red zones are detected, via SMS and dashboard push notification
- Support low-bandwidth mode: serve static PNG heatmap snapshot instead of interactive map when connection < 2G
- Accept data input from: mobile app submissions, SMS symptom reports, ASHA worker tablet entries, pharmacy dispensing logs

6.1.2 Data Privacy Requirements

- All heatmap data must be aggregated at the village level — no individual patient location or identity is ever displayed
- Minimum threshold of 5 cases required before a village appears on the heatmap (k-anonymity protection)
- Data retention: heatmap source records retained for 2 years; aggregated trend data retained indefinitely

6.1.3 Non-Functional Requirements

- Heatmap must load within 3 seconds on a 3G connection
- System must handle 10,000 concurrent heatmap viewers during outbreak events
- Offline mode: show last cached heatmap with a 'Last updated: X hours ago' timestamp

6.2 Ayushman Card-Linked Health Records

Patients can scan their Ayushman Bharat Health Account (ABHA) card using the app camera or enter their 14-digit ABHA ID to access their complete digital health record. The system fetches and locally caches all available records.

6.2.1 Record Fields Displayed

Category	Data Fields	Source
Identity	Name, DOB, Gender, Blood Group, ABHA ID	ABHA Registry API
Allergies	Drug allergies, food allergies, known reactions	PHC records + patient-reported
Vitals History	BP, blood oxygen (SpO2), body weight, BMI, temperature over time	Previous consultations + ASHA records
Lab Results	Haemoglobin (Hb), blood sugar (FBS/PPBS), lipid profile, urine routine	Linked lab results via ABHA
Chronic Diseases	Hypertension, diabetes, COPD, heart disease, TB, HIV status	Doctor-certified diagnoses
Medications	Current prescriptions with dose, frequency, duration	Pharmacy + Doctor records
Vaccinations	COVID-19, influenza, tetanus, hepatitis B, polio — dates and batch	COWIN integration
Visit History	All past teleconsult and in-person PHC visits with notes	Consultation records

6.2.2 AI Wellness Recommendations

Based on the patient's health record, the AI Wellness Engine generates personalised, actionable daily health recommendations. These are updated after each consultation and delivered in the patient's preferred language.

- Recommendations are evidence-based, generated by a fine-tuned GPT-4 model with a medical knowledge base
- Each recommendation includes a severity indicator: Informational, Advisory, or Urgent
- Categories covered: diet & nutrition, physical activity, medication adherence reminders, sleep hygiene, hydration, seasonal precautions
- For diabetic patients: daily blood sugar tracking reminders with trend alerts
- For hypertensive patients: daily BP log prompt with visual trend chart
- Recommendations are accessible as text, audio playback, and SMS summary
- ASHA workers receive aggregated risk scores for their assigned households to prioritise home visits

6.2.3 Lifestyle Risk Score

A 0–100 composite risk score is calculated from a lifestyle questionnaire (sleep quality, diet patterns, physical activity, stress levels, smoking/alcohol use, water source quality) and medical history. The score is recalculated monthly and displays a trend line. High-risk patients (score > 70) are flagged for proactive outreach by the ASHA worker.

6.3 Voice-Based Symptom Input with Multilingual NLP

Patients describe their symptoms verbally in their native language. The system transcribes, understands, and converts this into a structured symptom record for triage processing.

6.3.1 NLP Pipeline Specification

1. Patient presses the microphone button (or calls the IVR number for feature phones)
2. System prompts: 'Please describe what has happened to you' (in patient's detected/preferred language)
3. Speech is captured as raw audio and sent to the Speech-to-Text (STT) engine
4. STT Engine: Google Cloud Speech-to-Text with custom medical vocabulary model for Indian languages
5. Transcription is passed to the Medical NLP Model (custom fine-tuned BERT-based model trained on Indic medical corpora)

6. NLP model extracts: symptoms (named entity recognition), duration, severity descriptors, body parts, associated symptoms
7. Extracted entities are mapped to ICD-10 symptom codes for standardised processing
8. Structured symptom JSON is passed to the Triage Engine (see Section 6.5)
9. Patient hears a confirmation: 'I understood: [summary of symptoms]. Is this correct?' (Yes/No voice or keypad response)
10. If corrections needed, patient can re-record or edit individual fields via text

6.3.2 Languages Supported (v1.0)

- Hindi, Bengali, Telugu, Marathi, Tamil, Gujarati, Kannada, Odia, Punjabi, Malayalam
- Bhojpuri, Rajasthani, Maithili, Santali (feature phone IVR only for these dialects in v1.0)
- Language auto-detection using language identification model (no manual selection required)
- Confidence score displayed to operator; if < 80%, system requests re-recording

6.3.3 NLP Model Specifications

Model Architecture	Fine-tuned multilingual BERT (mBERT) + CRF layer for NER
Training Data	100,000+ anonymised PHC consultation transcripts + Indic medical text corpora
Task 1 — NER	Extract symptom entities, body parts, severity, duration from free text
Task 2 — Intent	Classify: new complaint / follow-up / medication query / emergency
Task 3 — Severity	Map extracted symptoms to a preliminary severity score (input to triage)
Accuracy Target	> 92% symptom extraction F1-score on validation set
Latency Target	< 2 seconds end-to-end for a 30-second audio input
Fallback	If NLP confidence < 70%, route to text-based questionnaire automatically

6.4 Appointment Booking — Online & Offline (SMS) Integration

6.4.1 Online Booking (App / Web)

- Patient selects available doctor and time slot from real-time availability calendar
- Slots display estimated wait time based on current queue depth and AI predictions
- Booking confirmation sent via push notification (app) and SMS
- Patients can reschedule or cancel up to 30 minutes before their slot
- Calendar syncs in real time — slot released immediately on cancellation

6.4.2 Offline SMS Booking Flow

For patients without internet access, the full booking flow is available via SMS:

Step	Patient SMS	System Response
1	Send START to 14441	Welcome! Reply with your ABHA ID or NEW to register
2	Send ABHA ID (e.g., 1234-5678-1234)	Hi Raman! Book appointment? Reply BOOK + date (e.g., BOOK 25APR)
3	BOOK 25APR	Available slots: 1) 10:00 AM 2) 2:00 PM. Reply slot number
4	Reply: 1	Confirmed! Token #12. Est. wait: 45 min. Reply STATUS to check queue anytime
Status check	STATUS	You are #8 in queue. Est. wait: 22 min. Queue updating in real time
Cancel	CANCEL	Your appointment cancelled. Slot released. Reply BOOK to rebook

6.4.3 USSD Booking (Zero-Data)

- Dial *HEALTH# (*43258#) from any mobile number — works without data plan
- USSD menu guides patient through: language selection → ABHA ID entry → slot selection → confirmation
- Confirmation message delivered as SMS after USSD session completes

6.4.4 Missed-Call Auto-Callback

- Patient gives a missed call to the toll-free IVR number (e.g., 1800-XXX-XXXX)
- System logs the caller ID and auto-triggers an outbound IVR call within 5 minutes
- IVR guides patient through symptom collection and booking in their detected language
- If IVR booking is not completed, an ASHA worker is assigned to follow up within 2 hours
- Works on any feature phone — zero cost to patient, zero data required

6.4.5 Emergency Button

- A prominent 'SOS / Emergency' button is visible on the Patient Portal home screen at all times
- Single tap triggers: (1) GPS-based ambulance dispatch request to nearest equipped facility, (2) SMS alert to family contact, (3) Auto-dial to 108 (National Ambulance Service)
- For SMS users: texting EMERGENCY to 14441 triggers the same ambulance dispatch flow

- Dispatched ambulance ETA displayed in real time on patient's screen
- Emergency cases are automatically assigned CRITICAL priority in the triage queue
- Doctor is immediately notified of the incoming emergency case on their dashboard

6.5 AI Triage System with Dynamic Re-Prioritisation

The triage system is the clinical heart of Sehat Saathi. It processes symptom data (from voice, SMS, or direct input) through a 3-stage questionnaire to compute a severity score and assign the patient a prioritised position in the queue.

6.5.1 Three-Stage Triage Questionnaire

Stage 1 — Emergency Screening (3 questions, 30 seconds)

Purpose: Immediate identification of life-threatening conditions. If any Stage 1 indicator is positive, patient is immediately assigned CRITICAL priority and the doctor is notified.

Q#	Question	Critical Indicators
Q1.1	Are you having trouble breathing or chest pain right now?	Yes → CRITICAL
Q1.2	Have you lost consciousness or are you unable to move a limb?	Yes → CRITICAL
Q1.3	Is there uncontrolled bleeding or severe injury?	Yes → CRITICAL + Ambulance prompt

Stage 2 — Severity Assessment (8–12 questions, 3–5 minutes)

Purpose: Determine the clinical severity of the presenting complaint. Questions are dynamically selected based on the symptoms identified in Stage 1 and the NLP intake.

Domain	Sample Questions	Scoring Weight
Pain intensity	Rate your pain 1–10 (or describe: mild / moderate / severe)	High (0.25)
Duration	How long have you had this symptom?	Medium (0.15)
Progression	Is it getting worse, staying the same, or improving?	High (0.20)
Associated symptoms	Any fever, vomiting, rash, or dizziness?	Medium (0.15)
Chronic conditions	Do you have diabetes, heart disease, or BP problems?	High (0.15)
Age / vulnerability	Age, pregnancy status, recent surgery or hospitalisation	Medium (0.10)

Stage 3 — Context & History (3–5 questions, 1–2 minutes)

Purpose: Contextual information for doctor briefing. Includes: recent travel, contact with sick individuals, medication currently taken, known allergies, and reason for visit (new complaint / follow-up).

6.5.2 Severity Scoring Algorithm

The final severity score (0–10) is a weighted composite of Stage 2 domain scores, adjusted by:

- Age/vulnerability multipliers (> 60 years: +10%, < 5 years: +15%, pregnancy: +20%)
- Chronic disease penalties (diabetes: +0.5, heart disease: +1.0, immunocompromised: +1.5)
- NLP-extracted symptom severity from voice input
- Temporal urgency: symptoms worsening rapidly → +1.0 bonus

Level	Score	Condition Examples	Queue Target	Doctor Notified?
CRITICAL	Score 8–10 | Chest pain, stroke, breathing difficulty, emergency trauma | Immediately | Yes — alert + audio			
URGENT	Score 5–7 | High fever, severe pain, persistent vomiting, suspected dengue | < 15 minutes | Yes — next in queue			
MODERATE	Score 3–4 | Mild infection, controlled chronic condition, joint pain | < 45 minutes | Normal queue			
ROUTINE	Score 1–2 | Cold, prescription renewal, wellness check, vaccination | < 90 minutes | Normal queue			

6.5.3 Dynamic Re-Prioritisation

- Re-scoring triggers: patient submits a symptom update while waiting, ASHA worker flags a change in condition, doctor elevates priority manually
- Re-prioritisation runs every 5 minutes automatically for all queued patients
- A patient whose score increases by 2+ points is moved up the queue immediately with notification to doctor
- Re-priority events are logged for audit trail and quality review

- If a patient's condition improves (score drops by 2+), they may be voluntarily moved down (with patient consent via SMS)

6.6 Predictive Queue Time & Transparency Feature

Patients can see exactly where they stand in the queue and understand why others are ahead of them. This builds trust, reduces frustration, and decreases consultation abandonment rates.

6.6.1 Patient Queue View

- Displays: current token number, position in queue, estimated wait time, and a simplified reason for each patient ahead (e.g., 'Higher severity case' — no personal data shown)
- Wait time prediction uses: current queue depth, historical doctor consultation duration by case type, time-of-day patterns, and live consultation duration tracking
- Patient receives SMS alerts at: 30 minutes before turn, 15 minutes before turn, and when they are next
- 'Why is this person ahead of me?' tap to expand shows severity category only (not personal details) — this is the trust-building mechanism

6.6.2 Predictive Queue Model

Algorithm	Gradient Boosting Regression (XGBoost)
Features	Queue depth, severity distribution, doctor's avg consultation time (rolling 7-day), time of day, day of week, patient age
Output	Predicted wait time in minutes with ±10 min confidence interval
Accuracy Target	< 15% mean absolute percentage error on wait time predictions
Update Frequency	Recalculated after every consultation completion

7. Doctor Portal — Feature Requirements

The Doctor Dashboard is a streamlined, information-dense interface designed for fast-paced PHC environments. Doctors access it via any web browser on laptop, tablet, or phone.

7.1 Pre-Consultation Patient Brief

Before each teleconsultation begins, the doctor sees an AI-generated structured patient brief (generated by GPT-4 from the patient's health record and triage questionnaire responses):

- Reason for visit (NLP-extracted from voice input, plain language summary)
- Current symptoms with severity indicators and duration
- Relevant health history: chronic conditions, recent diagnoses, allergies (drug allergies highlighted in red)
- Current medications with potential interaction flags
- Latest vitals from ABHA records
- Triage severity score with contributing factors
- Suggested differential diagnoses (AI-generated, for doctor reference only — not shown to patient)

7.2 Queue Management Dashboard

- Live queue sorted by severity with colour-coded rows (red/orange/yellow/green)
- Doctor can manually elevate or lower any patient's priority with a mandatory reason note (audited)
- Click to start consultation launches the video/audio call with patient automatically
- Consultation timer visible to ensure equitable time allocation
- Notes typed during consultation are auto-saved every 30 seconds

7.3 Prescription & Pharmacy Integration

- Doctor writes prescription in structured form: drug name (generic preferred), dose, frequency, duration, route, special instructions
- System checks drug-allergy interactions against patient record and flags conflicts before saving
- On saving prescription: auto-routed to patient's nearest pharmacy with real-time inventory check

- If medicine unavailable at nearest pharmacy: system finds next-nearest pharmacy with stock and shows ETA
- Patient receives SMS with pharmacy name, address, and medicine availability confirmation
- Monthly prescription patterns are aggregated to generate proactive stock orders for pharmacies

7.4 Post-Consultation Workflow

- Doctor adds diagnosis (ICD-10 code + plain language note), prescription, follow-up date, and referral if needed
- System generates simplified diagnosis summary in patient's preferred language
- Text-to-Speech converts summary to audio clip (delivered via WhatsApp or SMS)
- Follow-up appointment auto-scheduled and added to patient's queue with lower priority
- Consultation record synced to ABHA patient profile

8. Pharmacy Module — Analytics Dashboard & Integration

8.1 Pharmacy Analytics Dashboard

The pharmacy dashboard gives local pharmacists and PHC administrators visibility into prescription fulfilment, stock health, and government scheme performance.

Metric	Description	Recommended Chart	Update Frequency
Total Prescriptions Received	Count of prescriptions routed to this pharmacy from the Sehat Saathi system	Daily bar chart (30-day trend)	Real-time
Fulfillment Rate	% of prescriptions fully dispensed vs partially dispensed vs unfulfilled	Donut chart + daily trend line	Real-time
Most Dispensed Medicines	Ranked list of top 20 medicines by quantity dispensed (monthly)	Horizontal bar chart (sortable)	Daily refresh
Out-of-Stock Incidents	Number of times a prescription could not be filled due to zero stock, by medicine and date	Heat-map calendar (stock-out days highlighted)	Real-time
Stock Level Alerts	Medicines below reorder threshold (configurable per item)	Alert list with traffic-light indicators	Hourly
Govt. Scheme Reimbursements	Value of medicines dispensed under PM-JAY, PMJAY, state schemes — pending vs received	Stacked bar chart: pending vs received by month	Weekly batch
Average Fulfillment Time	Time from prescription received to patient collection (door-to-collection)	Line chart with target benchmark overlay	Daily
Cross-Pharmacy Requests	Number of inter-pharmacy medicine transfer requests sent/received with resolution rate	Sankey or simple two-bar comparison	Daily
Monthly Consumption Forecast	AI-predicted next-month consumption per medicine based on seasonal patterns and queue trends	Column chart with forecast overlay (dashed line)	Monthly auto-generate
Revenue by Source	Breakdown: private pay vs PM-JAY vs state scheme vs Sehat Saathi platform	Pie chart	Monthly

8.2 Doctor Prescription Integration

8.2.1 Real-Time Prescription Routing

- When a doctor finalises a prescription, it is instantly transmitted to the linked pharmacy via secure API
- Pharmacy receives a push notification on their dashboard with patient token number and prescription details
- Pharmacy confirms receipt and estimated preparation time (default: 30 minutes for standard prescriptions)
- Patient receives SMS: 'Your prescription has been sent to [Pharmacy Name]. It will be ready by [time]'
- If the required medicine is out of stock, the pharmacy can: (1) suggest a generic equivalent (doctor approval required), or (2) trigger a cross-pharmacy request

8.2.2 Cross-Pharmacy Medicine Sourcing

11. Pharmacy A receives prescription for Medicine X — out of stock
12. System queries inventory API of all pharmacies within 10 km radius for Medicine X availability
13. Pharmacy B (3.2 km away) confirms availability; system displays transfer request to Pharmacy A
14. Pharmacy A sends inter-pharmacy request — Pharmacy B confirms transfer and estimated delivery time
15. Patient is notified: 'Medicine available at [Pharmacy B] — pickup or delivery estimated by [time]'
16. Transfer logged in both pharmacies' dashboards; reimbursement split handled automatically

8.2.3 Proactive Monthly Stock Planning

- At the end of each month, the system analyses prescription patterns from the past 3 months
- AI forecasting model (time-series ARIMA + seasonal adjustment) predicts next month's medicine demand
- Pharmacy receives a 'Monthly Stock Forecast Report' 7 days before month end with: predicted demand by medicine, recommended order quantities, current stock levels, suggested reorder dates
- Pharmacist can review, adjust quantities, and submit the order directly to their supplier via the dashboard
- System cross-checks forecasts against PHC doctor's planned consultation calendar for additional signal

8.3 Medicine Availability Check for Patients

- Patients can search for a medicine from their prescription in the Patient Portal
- System displays: availability at nearest pharmacy, estimated wait time if being transferred, generic alternatives if applicable
- Integrated map shows nearest pharmacy with stock pinned on map with distance and route

9. Simplified Audio Diagnosis Explanation

After every consultation, the patient receives their diagnosis and care instructions in simple, accessible language — delivered as both text and a playable audio clip in their preferred language. This is the 'Explain in Simple Mode' feature.

9.1 Generation Pipeline

17. Doctor finalises diagnosis note and prescription in the Doctor Dashboard
18. System passes doctor's clinical notes to GPT-4 with a prompt: 'Rewrite this diagnosis for a patient with Class 8 education in [language]. Avoid medical jargon. Include: what is wrong, why it happened, what the treatment is, how to take medicines, when to come back'
19. Output is reviewed for safety — drug dosage instructions are double-checked against prescription data
20. Text is converted to audio using Google Cloud Text-to-Speech in the patient's language and regional accent
21. Audio clip (MP3, < 90 seconds) is sent via WhatsApp message if patient has WhatsApp, otherwise via SMS link
22. Audio is also available for replay in the Patient Portal under 'My Consultations'

9.2 Content Structure of Audio Explanation

- Greeting: 'Hello [Name], here is a summary of your consultation today with Dr. [Doctor Name]'
- Diagnosis: Plain language description of the condition (e.g., 'You have a throat infection caused by bacteria')
- Cause: Simple explanation (e.g., 'This can happen when the body's defences are low or from cold water')
- Treatment: Each medicine with: what it's for, how many to take, when to take (morning/evening/with food), and for how many days
- Warning signs: 'If you experience [symptoms], come back immediately or call 108'
- Follow-up: 'Please come back on [date] for a check-up'
- Closing in local dialect with culturally appropriate phrase

9.3 Accessibility

- Audio playback controls: play/pause, rewind 10 seconds, speed control (0.75x for slow comprehension)
- Full text transcript shown alongside audio for literate patients
- ASHA worker can access the same audio for a patient in their household roster to assist with comprehension
- Audio is stored for 12 months in the patient's health record

10. Core Data Model

10.1 Key Entities

Entity	Key Attributes	Relationships
Patient	patient_id, abha_id, phone, language_pref, village_id, risk_score	Has many Consultations, HealthRecords, Prescriptions
HealthRecord	record_id, patient_id, vitals (JSON), allergies (array), chronic_conditions	Belongs to Patient; updated per Consultation
Consultation	consult_id, patient_id, doctor_id, triage_score, status, notes, audio_url	Links Patient + Doctor; has Prescription
TriageSession	session_id, patient_id, q1_answers, q2_answers, q3_answers, score, severity	Belongs to Consultation
QueueEntry	entry_id, patient_id, doctor_id, position, score, created_at, updated_at	Belongs to Patient; ordered by score DESC
Prescription	rx_id, consult_id, doctor_id, items (JSON array), status, pharmacy_id	Links Consultation + Pharmacy
PharmacyItem	item_id, pharmacy_id, medicine_code, quantity, reorder_level, last_updated	Belongs to Pharmacy; linked to Prescription items
HeatmapPoint	point_id, village_id, symptom_type, count, severity_level, timestamp	Aggregated anonymised data; no patient link
Village	village_id, name, district, latitude, longitude, asha_worker_id	Linked to Patients, HeatmapPoints
Doctor	doctor_id, name, specialisation, languages, available_slots (JSON)	Has many Consultations, QueueEntries
Pharmacy	pharmacy_id, name, village_id, latitude, longitude, owner_phone, verified	Has many PharmacyItems, Prescriptions

11. Key API Contracts

11.1 Triage & Queue APIs

Endpoint	Method	Description	Response
POST /api/triage/start	POST	Initialise a triage session for a patient	{ session_id, questions: [ ] }
POST /api/triage/submit	POST	Submit answers; receive severity score	{ score, severity, next_steps }
GET /api/queue/{doctor_id}	GET	Fetch ordered queue for a doctor	{ entries: [{ position, patient_summary, score }] }
POST /api/queue/reprioritise	POST	Trigger re-score for all entries in a queue	{ updated_entries: [ ] }
GET /api/queue/eta/{entry_id}	GET	Get predicted wait time for a queue entry	{ wait_minutes, confidence_interval }
POST /api/emergency	POST	Trigger emergency — dispatch ambulance + notify doctor	{ ambulance_eta, doctor_notified: true }

11.2 Health Record APIs

Endpoint	Method	Description	Response
GET /api/records/{abha_id}	GET	Fetch patient health record from ABHA	{ vitals, allergies, medications, conditions }
POST /api/records/wellness	POST	Generate AI wellness recommendation for patient	{ recommendations: [ ], risk_score }
GET /api/records/history/{patient_id}	GET	Fetch full consultation history	{ consultations: [ ] }

11.3 Pharmacy APIs

Endpoint	Method	Description	Response
POST /api/pharmacy/prescription	POST	Route prescription to pharmacy	{ pharmacy_id, eta_minutes, confirmed }
GET /api/pharmacy/stock/{medicine}	GET	Check stock at all nearby pharmacies	{ pharmacies: [{ name, available, distance }] }

POST /api/pharmacy/cross-request	POST	Initiate cross-pharmacy medicine transfer	{ request_id, source_pharmacy, eta }
GET /api/pharmacy/forecast/{id}	GET	Get monthly stock forecast for pharmacy	{ forecasts: [{ medicine, predicted_qty, reorder_by }] }

12. Non-Functional Requirements

Category	Requirement	Target
Performance	API response time (p95)	< 500 ms for all non-AI endpoints
Performance	AI triage scoring latency	< 3 seconds end-to-end
Performance	NLP pipeline latency (30-sec audio)	< 2 seconds
Performance	Heatmap load time on 3G	< 3 seconds
Availability	System uptime SLA	99.5% monthly (< 3.6 hrs downtime/month)
Scalability	Concurrent active users	50,000 (target for 500 PHCs)
Offline	PWA offline functionality	Core features (queue check, record view) work offline
Security	Data encryption	AES-256 at rest; TLS 1.3 in transit
Security	Authentication	OTP-based login (no password required for patients)
Compliance	Health data standards	ABDM (ABHA) compliant; DPDP Act 2023 compliant
Accessibility	WCAG level	2.1 Level AA for web interfaces
Localisation	Languages at launch	10 languages (Hindi + 9 regional)
Device support	Minimum device spec	Android 5.0+; iOS 12+; any modern browser
SMS latency	SMS delivery time	< 30 seconds for appointment confirmations

13. Release Roadmap

Phase	Timeline	Deliverables
Phase 0 — Foundation	Month 1–2	Auth system, patient registration (app + SMS), ABHA integration, basic health record display
Phase 1 — Triage Core	Month 2–3	3-stage triage questionnaire, severity scoring, static queue, doctor dashboard v1
Phase 2 — NLP + Voice	Month 3–4	NLP model training + integration, voice input in Hindi + 3 regional languages, IVR flow
Phase 3 — Queue Intel	Month 4–5	Dynamic re-prioritisation, predictive wait time, queue transparency view, SMS alerts
Phase 4 — Pharmacy	Month 5–6	Pharmacy module, prescription routing, stock check, cross-pharmacy requests, analytics dashboard
Phase 5 — Heatmap	Month 6–7	Heatmap system, outbreak detection algorithms, DHO alert system, ASHA worker data entry
Phase 6 — Audio Diag	Month 7–8	Audio diagnosis generation, TTS integration, WhatsApp delivery, wellness AI recommendations
Phase 7 — Hardening	Month 8–9	Performance optimisation, security audit, ABDM certification, pilot with 5 PHCs
v1.0 Launch	Month 9–10	Full production launch across 50 PHCs in pilot district, with monitoring and feedback loops

14. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
NLP accuracy low for rare dialects	Medium	High	Fallback to text questionnaire; continuous model retraining pipeline with ASHA-collected data
Pharmacy inventory API non-compliance	High	Medium	SMS-based manual stock update as fallback; incentivise pharmacists with analytics value
PHC doctors reluctant to adopt new system	Medium	High	Champion doctor programme; 1-hour onboarding; minimal click redesign; offline capability
ABHA API downtime / data unavailability	Medium	Medium	Local cache of last-known health record; flag stale data clearly to doctor
Patient data privacy breach	Low	Critical	End-to-end encryption, RBAC, quarterly security audits, DPDP Act compliance
Queue gaming (patients claiming higher severity)	Medium	Medium	ASHA worker verification for high-priority queue claims; doctor override audit trail
Low SMS delivery in remote areas	Medium	Medium	Dual SMS provider (Twilio + BSNL gateway); 3 retry attempts; IVR fallback
Ambulance dispatch failure (emergency button)	Low	Critical	Direct dial to 108 as primary; system dispatch as secondary; regular integration testing

15. Success Metrics & KPIs

Metric	Baseline (Pre-launch)	6-Month Target	12-Month Target
Avg. patient wait time at PHC	3–4 hours	< 90 minutes	< 45 minutes
% critical cases seen within 15 min	Unknown (FIFO)	> 85%	> 95%
Patient registration completion rate	N/A	> 70%	> 85%
Triage accuracy (vs doctor rating)	N/A	> 85%	> 92%
Doctor pre-brief utilisation rate	0%	> 80% of consults	> 95%
Pharmacy prescription fulfillment rate	Unknown	> 75%	> 90%
Out-of-stock incidents per month	Unknown	< 20% reduction	< 40% reduction
Heatmap outbreak detection lead time	Reactive (2–4 weeks)	2–3 days	< 24 hours
Patient audio diagnosis listen rate	N/A	> 60%	> 75%
ASHA worker adoption rate	0%	> 60%	> 85%
System uptime	N/A	> 99.0%	> 99.5%

Appendix A — Glossary

Term	Definition
ABHA	Ayushman Bharat Health Account — India's national digital health identity
ABDM	Ayushman Bharat Digital Mission — the national digital health ecosystem framework
ASHA	Accredited Social Health Activist — community health workers at village level
DHO	District Health Officer — government official overseeing PHC network
IVR	Interactive Voice Response — automated phone system for voice menu navigation
NER	Named Entity Recognition — NLP task of extracting structured entities from text
PHC	Primary Health Centre — government-run rural health facility
PM-JAY	Pradhan Mantri Jan Arogya Yojana — government health insurance scheme
PMJAY	Pradhan Mantri Aarogya Mitra — health scheme field worker
PWA	Progressive Web App — web application with offline and native-like capabilities
SpO2	Peripheral oxygen saturation — measure of blood oxygen level
TTS	Text-to-Speech — AI system converting written text to spoken audio
USSD	Unstructured Supplementary Service Data — mobile menu system requiring zero data

Appendix B — References

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- Ayushman Bharat Digital Mission — ABDM Sandbox Documentation (abdm.gov.in)
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- Google Cloud Speech-to-Text — Medical Adaptation Guide
- BERT-based Multilingual Models for Indic NLP (AI4Bharat)
- Digital Personal Data Protection Act, 2023 — DPDP Compliance Guidelines

— End of Document —

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